

Lecture10: Probability Cheat Sheet

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Probability of event E:

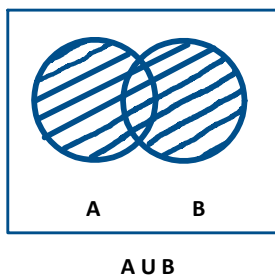
$$P(E) = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes in Sample Space}}$$

$$P(E) = \frac{|E|}{|S|}$$

1) Union of Events: Let we have two events A and B.

Union of events A and B contains all elements that are either in A or in B or in both A and B.

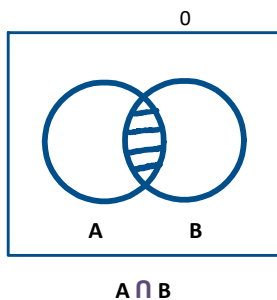
$$A \cup B = |A| + |B| - |A \cap B|$$



2) Intersection of Events: Let we have two events A and B.

Intersection of events A and B contains all elements that are in both A and B.

$$A \cap B = |A| + |B| - |A \cup B|$$



1) Mutually Exhaustive Events:

If union of all events cover all the elements in sample space then those events are called mutually exhaustive events.

If $A \cup B = S$ then A & B are mutually exhaustive events.

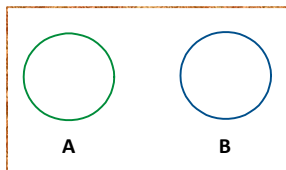
Example:

1. Tossing a coin
2. Throwing a die
3. Drawing a card

2) Mutually Exclusive Events:

If two events A & B cannot occur together
and intersection of two events A & B is zero
Then A & B are mutually exclusive events.

If $A \cap B = 0$ then A & B are mutually exhaustive events.



Event A & B are mutually Exclusive

Example:

1. Tossing a coin
2. Throwing a die
3. Drawing a card

1) Union and Intersection are :

1. Commutative:

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

2. Associative:

- $A \cup (B \cap C) = (A \cup B) \cap C$
- $A \cap (B \cup C) = (A \cap B) \cup C$

3. Distributive:

- $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
- $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

2) Compliments of Events ($\bar{E}$):

Complement of events consists of all elements
that are present in sample space (S)
but not present within event (E).

$\bar{E}$ occurs if E doesn't occur

$$\bar{E} = S - E$$

$$(|\bar{E}|/|S|) = (|S|/|S|) - (|E|/|S|)$$

$$P(\bar{E}) = 1 - P(E)$$

3) De Morgan's Law:

$$\bullet \overline{A \cap B} = \bar{A} \cap \bar{B}$$

3) De Morgan's Law:

- $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- $\overline{A \cap B} = \overline{A} \cup \overline{B}$

4) Subset:

Let we have two events E & F.

If all outcomes in E are also present in F

Then E is contained in F

We can say E is subset of F

$$E \subseteq F$$

5) Axioms of Probability:

- **Axiom 1:**
 $0 \leq P(E) \leq 1$
- **Axiom 2:**
 $P(S) = 1$
- **Axiom 3:**
For any sequence of mutually exclusive events
 $E_1, E_2, E_3, \dots, E_n$ ($E_i \cap E_j = \emptyset$ for all i, j)
 $P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n)$

6) Odds in Favour:

Ratio of no. of favourable outcomes to no. of unfavourable outcomes of a particular event is called odds in favour of an event.

$$\text{Odds in favour of event } A = P(A)/P(\overline{A}) = P(A)/(1-P(A)) = |A|/|\overline{A}|$$

Odds in Against:

Ratio of no. of unfavourable outcomes to no. of favourable outcomes of a particular event is called odds in against of an event.

$$\text{Odds in against of event } A = P(\overline{A})/P(A) = 1-P(A)/P(A) = |\overline{A}|/|A|$$

If E & F are two events,

Conditional Probability of event E

Given that event F has already happened

Is denoted by $P(E|F)$

$$P(E|F) = P(E \cap F) / P(F)$$

Multiplication Theorem:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

1. Independent Event:

If $P(A|B) = P(A)$

Then we can say A is independent of B.

Occurring of B does not affect the occurring of A.

If $P(A|B) = P(A)$

Then $\frac{P(A \cap B)}{P(B)} = P(A)$

$$P(A \cap B) = P(A) \cdot P(B)$$

Then A & B are independent.

If two events A & B are not independent then they are dependent.

- If A & B are independent then,

A & B are independent,
A & B are independent,
A & B are also independent.

- A is independent of any event

If $P(A) = 1$
& $P(A) = 0$

- Mutually exclusive events with non zero probability are always dependent.

If A & B are mutually exclusive
 $A \cap B = 0$

And if $P(A) \neq 0$ & $P(B) \neq 0$
Then $P(A \cap B) \neq P(A) \cdot P(B)$

Then A & B are dependent events.

Independence of Three events:

A, B & C are independent events if

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(B \cap C) = P(B) \cdot P(C)$$

$$P(C \cap A) = P(C) \cdot P(A)$$

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

1. Random Variable (X):

Random variable is a mathematical rule or a function that assigns a numerical value to the outcome of a random experiment.

Example: Random variable, X = Number appearing when I throw a die.

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$P(X=1) = 1/6$$

$$P(X=2) = 1/6$$

$$P(X=3) = 1/6$$

$$P(X=4) = 1/6$$

$$P(X=5) = 1/6$$

$$P(X=6) = 1/6$$

Probability of a random variable: Probability of an event E of a random variable X is denoted as $P(X = E)$ or $P(E)$

For any random variable, sum of probability of all outcome in sample space is equal to 1.

If two random variables are dependent in that case there are three types of probability we need to consider those are:

2. Joint Probability
3. Marginal Probability
4. Conditional Probability

Random Experiment: Throwing a die and tossing a coin

X : Number appearing on a die; Possible values: $\{1, 2, 3, 4, 5, 6\}$

Y : Outcomes appearing on a toss; Possible values: $\{H, T\}$

1. Joint Probability:

Likelihood of two events happening simultaneously

$$P(A \cap B) \text{ or } P(X = A \text{ and } Y = B)$$

$$P(X = 3 \text{ and } Y = H)$$

$$= P(X = 3 \cap Y = H)$$

$$= 1/6 \times 1/2$$

$$= 1/12 \text{ (Though } X \text{ \& } Y \text{ are independent)}$$

Sum of all possible values in a joint probability distribution is equal to 1.

Joint probability distribution table:

X/Y	X	Y
1	1/12	1/12
2	1/12	1/12
3	1/12	1/12
4	1/12	1/12
5	1/12	1/12
6	1/12	1/12

1. Marginal Probability:

Probability of a single event without considering the occurrence of other events.

It is obtained by summing the joint probabilities over the other events.

Marginal probability of getting a 3 on the die is

$$P(X = 3) = P(X = 3, Y = H) + P(X = 3, Y = T)$$

$$= 1/12 + 1/12$$

$$= 2/12$$

$$= 1/6$$

Marginal probability of getting a Head is

$$P(X = H) = P(X = 1 \cap Y = H) + P(X = 2 \cap Y = H) + P(X = 3 \cap Y = H) + \dots + P(X = 6 \cap Y = H)$$

$$= 1/12 + 1/12 + 1/12 + 1/12 + 1/12 + 1/12$$

$$= 6/12$$

$$= 1/2$$

X/Y	H	T
1	1/12	1/12
2	1/12	1/12
3	1/12	1/12
4	1/12	1/12
5	1/12	1/12
6	1/12	1/12

4. Conditional Probability:

probability of an event happening given that another event has already occurred.
It is denoted as $P(X | Y)$

Conditional probability of getting a Head given that the die shows at 3 is

$$P(Y = H | X = 3) = \frac{P(Y = H, X = 3)}{P(X = 3)}$$

$$= \frac{P(Y = H, X = 3)}{P(X = 3, Y = H) + (X = 3, Y = T)} = \frac{1/12}{1/6} = \frac{1}{2} \quad \begin{array}{l} \text{Joint Probability} \\ \text{Marginal Probability} \end{array}$$

Though tossing a coin is an independent event and its probability is always $1/2$

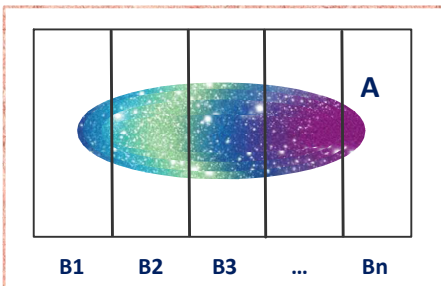
Conditional Probability = Joint Probability / Marginal Probability

1. Total Probability:

Total probability is the way of relating marginal probabilities to conditional probabilities

Total probability of an event is expressed with multiple distinct events.

We have to find the probability of event A; but the sample space B is divided into multiple parts exactly like shown in the figure.



$$B = \{B1 \cup B2 \cup B3 \dots \cup Bn\}$$

Here sample space B is set of pairwise disjoint events whose union is entire sample space

Here probability of any event $P(A) = \sum P(A \cap Bn)$

$$P(A) = \sum P(A | Bn) \cdot P(Bn)$$

$$P(A) = P(A \cap B1) + P(A \cap B2) + P(A \cap B3) + \dots + P(A \cap Bn)$$

$$P(A) = P(A | B1) \cdot P(B1) + P(A | B2) \cdot P(B2) + P(A | B3) \cdot P(B3) + \dots + P(A | Bn) \cdot P(Bn)$$

Experiment 1: $P(\text{Rain}) = 60\%$

$P(\text{Not Rain}) = 40\%$

$P(\text{School} | \text{Rain}) = 10\%$

$P(\text{School} | \text{Not Rain}) = 80\%$

$P(\text{School}) = ?$

By applying Total Probability Formula
Probability of going to school is:

$$P(\text{School} | \text{Rain}) \cdot P(\text{Rain}) + P(\text{School} | \text{Not Rain}) \cdot P(\text{Not Rain})$$

$$= 0.10 \times 0.60 + 0.80 \times 0.40$$

$$= 0.06 + 0.32$$

$$= 0.38$$

$P(\text{School}) = 38\%$

According to conditional probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

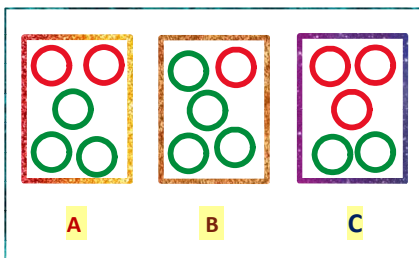
According to multiplication theorem:

$$P(A \cap B) = P(B | A) \cdot P(A)$$

Now we can write:

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

This is the expression of Bayes Theorem



Q. Pick a bag at random
And draw a random ball out of it.

1. Probability of getting a red ball given bag a is chosen ?

Conditional Probability

$$P(\text{Red} | A) = 2/5$$

2. Probability of getting a red ball and choosing bag A ?

Multiplication Theorem

$$P(\text{Red} \cap A) = P(\text{Red} | A) \cdot P(A)$$

$$= 2/5 \cdot 1/3$$

$$= 2/15$$

3. Probability of getting a red ball ?

Total Probability

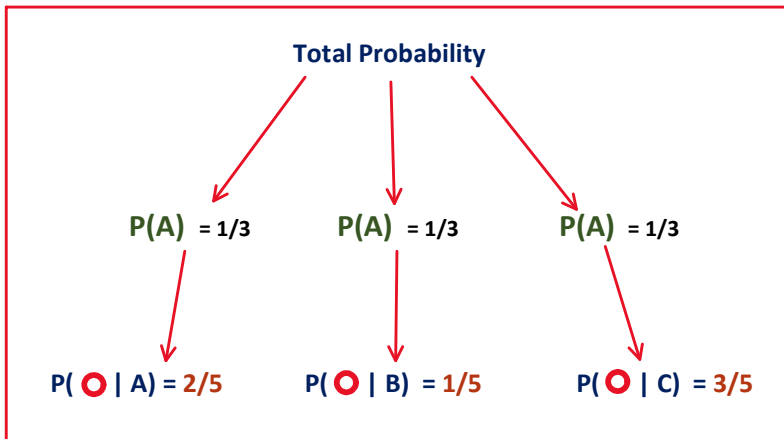
3. Probability of getting a red ball ?

Total Probability

$$P(\text{Red}) = P(\text{Red} | A) \cdot P(A) + P(\text{Red} | B) \cdot P(B) + P(\text{Red} | C) \cdot P(C)$$

$$= \frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3}$$

$$= \frac{2}{5}$$



4. Probability of bag a is chosen given that the drawn ball is red ?

Total Probability

$$P(\text{Red}) = P(\text{Red} | A) \cdot P(A) + P(\text{Red} | B) \cdot P(B) + P(\text{Red} | C) \cdot P(C)$$

Contribution of
Bag A

Contribution of
Bag B

Contribution of
Bag C

$$P(A | \text{Red}) = \frac{P(\text{Red} \cap A)}{P(\text{Red})} = \frac{\text{Contribution of Bag A}}{\text{Total Probability of red ball}}$$

$$P(A | \text{Red}) = \frac{\frac{2}{5} \cdot \frac{1}{3}}{\frac{2}{5} \cdot \frac{1}{3} + \frac{1}{5} \cdot \frac{1}{3} + \frac{3}{5} \cdot \frac{1}{3}}$$

$$P(A | \text{Red}) = \frac{\frac{2}{15}}{\frac{2}{5}} = \frac{1}{3}$$

Q2. Suppose two factories supply light bulbs to the market

- Factory A's bulb work for over 5000 hours 99% of the cases.
- Factory B's bulb work for over 5000 hours 95% of the cases.

Factory A supplies 60% of the total bulbs.

What is the chance that a randomly purchased bulb will work for more than 5000 hours ?

Let event E = randomly purchased bulb will work for more than 5000 hours

Total Probability

$$P(E) = P(E | A) \cdot P(A) + P(E | B) \cdot P(B)$$

$$P(E) = 0.99 \cdot 0.6 + 0.95 \cdot 0.4$$

$$P(E) = 0.97$$

Given that -

The purchased bulb worked for more than 5000 hours.

What is the probability that it was produced by factory A?

Bayes Theorem

$$P(A | E) = \frac{\text{Contribution of Factory A}}{\text{Total Probability of E}} = \frac{P(E | A) \cdot P(A)}{P(E | A) \cdot P(A) + P(E | B) \cdot P(B)}$$

$$P(A | E) = (0.99 \times 0.6) / (0.97) \\ = 0.6124$$

probability that it was produced by factory A is 61 %